

Wai Tong Chung

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Experience

Together AI

AI Researcher

San Francisco, CA

July 2024 - Present

- Investigating pre- and post-training methods for language models in inference optimization and agentic applications.
- Developed speculator model training methods that resulted in the fastest B200 inference of DeepSeek-R1 (July 2025).
- Contributed >10K lines of code to a Kubernetes-based project for automating org-wide training/inference experiments.
- Authored 4 technical blogs on reasoning model inference optimization through training speculator models.

Stanford University

Machine Learning Graduate Research Assistant

Stanford, CA

Sept. 2018 - June 2024

- Investigated and developed deep learning methods for efficient high-performance computing software in flow physics.
- Authored 20+ AI for science and computational engineering refereed publications in top ML and engineering venues.
- Contributed significantly to accepted NSF, NASA, U.S. DoE, and Google grant proposals (total worth > \$1.5M).

JPMorgan Chase & Co.

Financial Messaging Software Engineer

United Kingdom

Sept. 2017 - Aug. 2018

- Developed, tested, and deployed a Java-based financial messaging application that processed \$6T of daily payments.

Education

Stanford University

Ph.D. in Mechanical Engineering.

Stanford, CA

Sept. 2018 - June 2024

Advised by Prof. Matthias Ihme in Flow Physics and Computational Engineering Group.

Thesis: Overcoming Small Datasets in Machine Learning Studies of Multi-Physics Flows in Propulsion. [\[url\]](#)

Research Focus: AI for Science, High-Performance Computing, Scientific Machine Learning.

Imperial College London

B.Eng. M.Eng. in Mechanical Engineering with First Class Honours.

United Kingdom

Sept. 2013 - Aug. 2017

Awards: Most Outstanding Thesis (Top 1 of 138 students), Dean's List (Top 10% of 138 students).

Course Focus: Computational Science, Flow Physics, Statistical Mechanics, Linear Algebra, Probability Theory.

Selected Writing

See [Google Scholar](#) for full list of academic publications.

Technical Blogs

W.T. Chung, D. Waters, A. May, B. Athiwaratkun. Boosting DeepSeek-R1's Speed with Customized Speculative Decoding. *Together AI Blog*, 2025. [\[url\]](#)

S. Zhu, F. Bianchi, **W.T. Chung**, Z. Hasan, R. Wu, C. Zhang, J. Zou, B. Athiwaratkun. How Together AI Uses AI Agents to Automate Complex Engineering Tasks: Lessons from Developing Efficient LLM Inference Systems. *Together AI Blog*, 2025. [\[url\]](#)

J. Wang, **W.T. Chung** et al. AdapTive-LeArning Speculator System (ATLAS): A New Paradigm in LLM Inference via Runtime-Learning Accelerators. *Together AI Blog*, 2025. [\[url\]](#), [\[press\]](#)

Together AI[†]. Together AI Delivers Top Speeds for DeepSeek-R1-0528 Inference on NVIDIA Blackwell. *Together AI Blog*, 2025. ([†]**W.T. Chung** contributed to speculative decoding, benchmarking, as well as blogpost drafting/editing). [\[url\]](#)

Refereed Journal, Conference, and Workshop Articles

M Ihme[†], **W.T. Chung**[†]. Artificial Intelligence as a Catalyst for Combustion Science and Engineering[‡]. Accepted in *Proc. Combust. Inst.* 40, 2024. ([†]Equal contribution. [‡]Presented as a plenary lecture at the 40th International Symposium on Combustion, Milan, 2024 [\[.pdf\]](#))

W.T. Chung *et al.* Turbulence in Focus: Benchmarking Scaling Behavior of 3D Volumetric Super-Resolution with BLASTNet 2.0 Data. *Adv. Neural Inf. Process. Syst. (NeurIPS)* 36, 2023. [[pdf](#), [press](#)]

M. Ihme[†], **W.T. Chung**[†], A.A. Mishra[†]. Combustion Machine Learning: Principles, Progress, and Prospects. *Prog. Energy Combust. Sci.* 91:101010, 2022. ([†]*Equal contribution*) [[pdf](#)]

W.T. Chung, K.S. Jung, J. H. Chen, M. Ihme. The Bearable Lightness of Big Data: Towards Massive Public Datasets in Scientific Machine Learning. In: *ICML AI4Science Workshop*, 2022. [[pdf](#)]

D.D. Wu, **W.T. Chung**, M. Ihme. ML for Safely Landing on Mars. In: *NeurIPS ML4PS Workshop*, 2022. [[pdf](#)]

W.T. Chung, A.A. Mishra, N. Perakis, M. Ihme. Accelerating High-fidelity Combustion Simulations with Classification Algorithms. In: *AAAI MLPS Spring Symp.*, 2021. [[pdf](#), [video](#)]

W.T. Chung, A.A. Mishra, N. Perakis, M. Ihme. Random Forests for Accelerating Turbulent Combustion Simulations. In: *NeurIPS ML4PS Workshop*, 2020. [[pdf](#)]

Accepted Grant Proposals

NSF Pathways to Enable Open-Source Ecosystems Grant (Awarded \$1.2M). PI: M. Ihme, 2024. [[info](#)]

Google Award for Inclusion Research Grant (Awarded \$60K). PI: M. Ihme, 2022. [[info](#)]

US Department of Energy NERSC Grant (Awarded 11.2M core-hours). PI: M. Ihme, 2022. [[info](#)]

NASA Early Stage Innovations Grant (Awarded \$650K). PI: M. Ihme, 2021. [[info](#)]

Honors and Awards

Stanford CS323: The AI Awakening Best Final Project Prize (Top 4 of 87 Students)	2023
Stanford Human-Centered AI Affinity Group Award [info , press]	2023
Stanford Human-Centered AI Graduate Fellowship [info , press]	2022-2023
Stanford School of Engineering Graduate Fellowship	2018-2019
Imperial College Mechanical Engineering Most Outstanding Thesis Prize (Top 1 of 138 Students)	2017
Imperial College Mechanical Engineering Dean's List (Top 10% of 138 Students)	2017

Selected Professional Activities

Lead Organizer for *Future Learning Approaches for Modeling and Engineering (FLAME) AI Workshop*, 2023. [[info](#)]

Lead Organizer for *Stanford HAI Climate-Centered AI Seminar Series*, 2023. [[press](#)]

Affiliate for *Stanford Data Science Center for Open and REproducible Science*, 2023-2024.

Reviewer: *ICLR* 2025; *AISTATS* 2025; *NeurIPS*, 2024, 2025; *ML and the Physical Sciences Workshop at NeurIPS*, 2021, 2022, 2023, 2024; *Synergy of Scientific and Machine Learning Modeling Workshop at ICML*, 2023; *ReScience C (ML Reproducibility Challenge)*, 2023; *AI for Science: Progress and Promises Workshop at NeurIPS*, 2022, 2024; *Proceedings of the Combustion Institute*, 2024; *Signal, Image and Video Processing*, 2024; *ASME Turbomachinery Technical Conference & Exposition*, 2023; *Combustion and Flame*, 2023, 2024; *International Journal of Engine Research*, 2023.

Skills

Programming and Engineering

Proficient: Python, PyTorch, Slurm, Docker

Familiar: C++, Kubernetes, Ray, MPI, TensorFlow, MATLAB, FORTRAN, Java

Languages

Proficient: English, Malay.

Familiar: Mandarin, Cantonese.